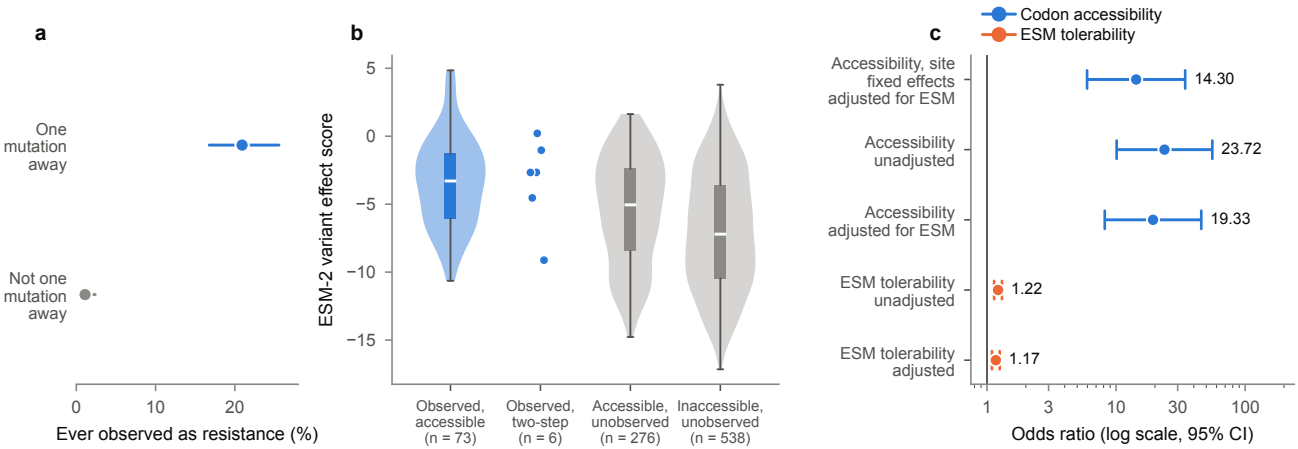
a
One mutation away
Not one mutation away
0
10
20
Ever observed as resistance (%)
b
ESM-2 variant effect score
5
0
−5
−10
−15
Observed, accessible (n = 73)
Observed, two-step (n = 6)
Accessible, unobserved (n = 276)
Inaccessible, unobserved (n = 538)
c
Codon accessibility
ESM tolerability
Accessibility, site fixed effects adjusted for ESM
14.30
Accessibility unadjusted
23.72
Accessibility adjusted for ESM
19.33
ESM tolerability unadjusted
1.22
ESM tolerability adjusted
1.17
1
3
10
30
100
Odds ratio (log scale, 95% CI)